

TRELLIS

AI-guided predictive modeling for bioprocess development

Product Overview

What Trellis Is

Trellis is a product that helps bioprocess development teams rebuild their workflows around predictive models. It is built for the biopharma cell-culture process development stage — the step between discovering a promising molecule and being able to manufacture it reliably, repeatedly, and at scale.

The core idea: instead of relying on a person to manually design a small number of experiments and interpret the results, Trellis lets a system build a predictive model of the process, use that model to virtually explore the full design space, and recommend the next experiments most likely to reveal the optimum — while a human bioprocess scientist stays in control of the loop throughout.

The Problem It Solves

A therapeutic molecule only becomes a medicine if the manufacturing process — typically a cell-culture process — can produce it reliably across batches, scales, and sites. Two structural problems make this hard:

- Enormous design space: parameters such as media composition (20+ amino acids and sugar types), feed strategy, pH, temperature, aeration, and agitation can be combined in thousands of ways. A human can only reason about roughly three to four parameters at once, so most of the space goes unexplored and the true optimum can be missed.
- Batch-to-batch variation: every process parameter has an uncertainty band around its set point — no parameter can be held at an exact value. If a process sits on a narrow, sharp optimum rather than a wide, stable plateau, small natural drift in these bands can push output outside acceptable quality. Characterizing the process across its full operating range is also a regulatory requirement in biopharma.

The manual, heuristic version of this work is slow: a single experiment run typically takes about 15 days, plus another two to three weeks to analyze results and plan the next round. A handful of iterations to properly optimize a process can take a year to a year and a half — and delays on one molecule can back up others in a company's pipeline.

The New Paradigm

Rather than running one experiment path toward a single optimum, Trellis is built around this loop:

- Design experiments chosen to maximize information ("informative experiments"), not just to test a hypothesis.
- Feed results into a predictive model that learns process behavior (e.g., growth curves, substrate consumption curves).
- Use the trained model to simulate thousands of virtual conditions across the design space — far more than could be tested physically.

- Identify the operating region (the optimum), the robust region (safe parameter ranges around it), and the validated domain (the full range actually tested) — the last of which matters for regulatory submissions, since operating ranges must sit safely inside tested ranges.
- Run real lab experiments at the recommended conditions, feed results back into the model, and repeat until the space is adequately mapped.

The team was explicit that this is not a one-shot exercise: one experiment can never generate enough evidence to build a reliable model. The value comes from structured iteration — starting from a human-designed set of initial experiments, then letting the model fill in the gaps faster and across far more parameters (potentially 20–30 at once) than a person or classical statistical method could track.

How It Works: The Agent Architecture

Trellis keeps the bioprocess scientist in control, but gives them a set of specialized agents that support each stage of the loop. Deterministic tools handle the actual computation — the agents orchestrate and interpret, they do not perform backend calculations themselves, to avoid hallucination risk.

Agent	Role
Design agent	Works with the scientist to design the next set of experiments.
Protocol agent	Generates lab protocols matched to the fermenters and sampling times in use, and produces a data sheet for the lab scientist to record results on.
Data engineering agent	Converts results from each lab's own data format into a structured format the modeling agent can use.
Modeling agent	Builds candidate model architectures from the structured data, runs train/test splits, checks prediction accuracy, and selects the best-fitting architecture. Built on rules developed by a chemical engineering PhD with 7+ years of domain experience; the user can steer it or let it run automatically.
Optimization agent	Deploys the trained model across the design space to find where the optimum lies, incentivized against a target metric (e.g. titer). Stops on a convergence criterion (e.g. negligible improvement over recent steps, or a step budget) and hands the recommended conditions back to the scientist for real-world testing.

One loop = one round of virtual optimization ending in a real-world experiment recommendation. How many loops a bioprocess scientist runs — and when they stop — is entirely up to them, based on the improvement they need, their budget, and how many trials they can run. The team was careful to distinguish this from open-ended "loop engineering": each loop terminates after simulation, a human runs the physical experiment, and only then does the next loop begin.

Memory

The agentic framework retains memory at two levels, intended to avoid restarting from scratch on related work:

- Model memory — retained through the weights of the predictive models, e.g. carrying learned growth kinetics as a process scales from 1 liter to 20,000 liters within a project.
- Agentic (LLM) memory — retained across a project's lifecycle and, optionally, across a company's different projects, e.g. what lab equipment or analytical instruments a team has. This is intentionally lighter-weight than model memory and does not carry core process knowledge across organizations.

Memory retention is configurable: teams can turn it on or off, and cross-organization data boundaries can be enforced so that one customer's work never informs another's. The team flagged that the messaging around memory and cross-project retention is still unsettled internally and needs further clarification — including whether "we" in this context means the Trellis team or the system acting autonomously.

What Customers Are Actually Paying For

Two benefits accrue in parallel, but they are not equally important to the pitch:

- Time and experiments saved — a downstream effect of faster iteration.
- Increased certainty of reaching a viable process within budget — identified in the discussion as the primary thing customers pay for.

The reasoning: companies typically budget about a year for process development, but timelines often slip to a year and a half, which delays not just one molecule but other molecules queued behind it in the pipeline. Framing the product around "how much more certain are you of hitting your target in the time you've budgeted" (e.g., "Trellis increases that certainty by 90%") was proposed as a stronger opening hook than leading with the AI capability itself.

Example Application

Cell-culture processes of this kind are used to manufacture products such as monoclonal antibodies and insulin. A representative use case:

- Identify which of many process parameters matter most, via virtual experiments.
- Predict culture behavior (e.g., growth and substrate-consumption curves) under changes to those parameters.
- Once predictive quality is established, run large numbers of virtual experiments to narrow down the candidate region before committing to real lab work.

In practice, teams work with roughly tens of meaningfully distinct parameters (not hundreds) — for example, ~20+ amino acid and sugar components within media composition, plus feed-timing strategy — and the combinations of these run into the thousands. Without a model, experimenters typically vary only three to four factors at a time; Trellis is designed to vary all of them simultaneously, provided the model has learned each factor's effect from prior data.

Origin of the Name

"Trellis" started as an internal placeholder name, arrived at from a shortlist that included "tune," "thrust," and "torch." It stuck because a trellis is the wooden structure that supports and guides a growing plant — seen as a fitting metaphor for the product's role. As of this discussion, the name has not gone public; it has only been shared with one or two close contacts. The team likes the name but flagged the name and its origin story as open for revisiting before wider release.